

Dear Friend: Alleviating Anxiety through Co-creative Virtual Agent in Dual-Atmosphere Environments

Wenbo Jia
Beijing Normal University
China, China
1045846168@qq.com



Figure 1: Alleviating Anxiety through Co-creative Virtual Agent in Dual-Atmosphere Environments.

Abstract

This paper introduces a VR system for precise anxiety intervention, integrating LLM, Generative AI, and Unreal Engine 5.5 on Quest 3. Unlike generic apps, it employs a semi-automated pipeline to construct personalized 3D environments from user interviews, featuring a conversational virtual friend for support. A 6-user pre-experience indicates intervention effectiveness correlates strongly with anxiety type. For specific phobias in realistic exposure scenes, the virtual friend acted as a vital emotional anchor, significantly outperforming traditional relaxation. Conversely, for general anxiety, interactions provided improvement but lacked deep dependency induction. These findings suggest that virtual social companionship holds significantly greater therapeutic weight in high-stress contexts, offering key insights for future adaptive VR psychotherapy.

Keywords

Virtual Reality, Generative AI, Co-creative Agent, Anxiety

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1 Introduction

Anxiety disorders have become one of the most prevalent mental health issues globally. The latest data from the World Health Organization (WHO) [5] shows that over 1 billion people worldwide suffer from mental health disorders, with anxiety and depression being widespread among the youth population. Academic pressure, economic uncertainty, and information overload brought by digital life have led to a significant increase in the incidence of Generalized Anxiety Disorder (GAD) and specific phobias.

On one hand, although significant medical resources have been invested, there is still a need to seek more personalized and effective precision treatments within cognitive behavioral therapy [3].



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With breakthrough progress in Large Language Models (LLM) and Generative Artificial Intelligence [1], the participation of virtual reality in psychological healing has become more personalized and intelligent, addressing the pain points of traditional VR healing such as lack of personalization, high costs, and slow progress [4]. On the other hand, in virtual environments, audiences occasionally experience panic due to being alone in the virtual world, lacking an effective social support system. Although the social buffering effect in the field of social neuroscience has long confirmed that the presence of peers can significantly inhibit amygdala activation and physiological stress when an individual faces threats, the application of this mechanism in VR therapeutic environments is still in its nascent stage [2].

This study attempts to construct a dual-atmosphere experimental framework through users' active involvement and disclosure of details in preliminary interviews, where they personally conceive and design the image and traits of a fantasy friend, which is then synthesized and implanted into VR via Generative AI. The aim is to answer a core scientific question: How does the therapeutic function of a virtual partner generated under a preset co-creation mechanism dynamically shift with changes in environmental stress? Specifically, we hypothesize that in a high-fidelity exposure environment, this friend, who carries the user's projections, will act as a key emotional anchor, helping patients with specific phobias tolerate high-intensity stress responses through social buffering mechanisms; while in an abstract relaxation environment, it transforms into a gentle companionship resource, primarily assisting patients with general anxiety in emotional regulation.

2 Proposed method

2.1 Interview-based Preset Co-creation and GenAI VR System Construction

To explore the social buffering mechanism of virtual partners in different stress situations, this study constructed a semi-automated VR generation pipeline named "Dear Frien". This system abandons real-time generative dialogue, which carries higher clinical risks, and instead adopts an interview-based preset co-creation strategy to ensure the safety and controllability of the intervention process. Developed based on the Unreal Engine 5.5 engine, the system fully utilizes its Lumen global illumination and Nanite virtualized geometry technology to achieve ultimate visual immersion on Meta Quest 3.

The system construction consists of two core phases. First is the Co-creative Asset Generation Phase. Participants first describe in detail the "Safe House" where they feel most comfortable and the "Stress" features that terrify them the most during an interview. These qualitative descriptions are converted into specific prompts, input into Meta's WorldGen, and through steps such as LLM parsing, 3D models are generated and imported into UE5.5 as the basis for environmental lighting and background. Meanwhile, for the virtual partner, we use MetaHuman Creator combined with Generative AI-assisted texturing tools to rapidly generate a Virtual Agent based on the user's description of their "fantasy friend" or sketches drawn by hand. Utilizing the "IKEA effect" from psychology, the user's active investment establishes a foundation of psychological ownership and attachment to the virtual partner in advance.

During the interactive experience phase, the system establishes a complete intervention pipeline, ranging from initial user interviews to the construction of a dedicated virtual agent via the generative AI core. To clearly illustrate how the research is conducted across different contexts, Figure 4 details the two core experimental condition branches included within the system. For Specific Phobia, the system provides a High Arousal Stress exposure environment; in this context, relying on Unreal Engine 5's Behavior Tree and AI Perception System, the virtual companion—although lacking open-ended conversational capabilities—can intelligently trigger preset cognitive reappraisal voice prompts and comforting gestures based on the user's gaze and distance from fear sources (e.g., dark environments). Conversely, for Generalized Anxiety Disorder (GAD), the system offers a Low Arousal Stress comfortable and relaxing environment, utilizing the same underlying technology to drive the virtual character to provide guided relaxation and silent companionship. This design of physical presence and immediate responsiveness aims to simulate stable social companionship, enabling the virtual agent to serve precisely as the user's emotional anchor across different symptom types and environmental stress levels, ultimately leading to the therapeutic goal of a significant reduction in anxiety levels.

2.2 Experimental Design

To This study adopts a "symptom tendency type $\times$ environmental stress level" mixed design to verify the emotion regulation potential of virtual companions in dual-atmosphere VR environments through a preliminary experiment.

Participants&Grouping: Six youth participants (aged 18–30) were recruited. Based on baseline semi-structured interviews and the Trait subscale of the State-Trait Anxiety Inventory (STAI-T), participants were categorized into a Specific Phobia Tendency group ($N = 3$) and a Generalized Anxiety Disorder (GAD) Tendency group ($N = 3$). This grouping serves to explore mechanisms and present individual differences, rather than for between-group significance testing.

Experimental Task: Wearing a Meta Quest3, participants experienced a low-arousal comfortable environment and a high-arousal stressful environment in a counterbalanced order (~ 15 mins each). Both environments featured the same co-created virtual companion. Based on user behaviors like gaze and distance from stress cues, the system triggered the companion to provide comforting gestures, breathing prompts, and cognitive reappraisal audio, while synchronously recording key event timestamps.

Subjective Assessment: The STAI State subscale (STAI-S) and the Positive and Negative Affect Schedule (PANAS) were administered before and after each session to capture immediate changes in anxiety levels and shifts in emotional valence.

Physiological Assessment: A BIOPAC MP160 system (1000 Hz sampling rate) synchronously recorded and processed two signals: Electrodermal Activity (EDA): Mean Skin Conductance Level (SCL) and Skin Conductance Response (SCR) frequency were extracted to reflect the tonic baseline arousal and phasic rapid stress responses, respectively. Photoplethysmography (PPG): Filtered with a 0.5–5 Hz bandpass to extract Heart Rate Variability (HRV) metrics in both the time domain (SDNN, RMSSD) and frequency domain



Figure 2: UE5.5: Two types of virtual scene setup.



Figure 3: User interacting with Virtual Agent in both environment.

(LF, HF, LF/HF ratio). Together with EDA, HRV provides a complementary measurement of neuromodulatory balance and arousal intensity within the autonomic nervous system.

3 Evaluation

Based on the controlled preliminary experimental data from 6 users, qualitative and quantitative analyses revealed significant between-group interaction differences, strongly demonstrating that the efficacy of the intervention strategy is highly correlated with the participants' anxiety traits:

Specific Phobia Tendency Group ($N = 3$): In the high-arousal stressful environment (highly replicating personal fear triggers), the virtual companion was not viewed as a mere background character but acted as a crucial "emotional anchor." Physiological monitoring data (EDA and HRV) showed that when users faced fear sources leading to a surge in physiological arousal, the companion's intervention facilitated a rapid return of physiological metrics toward the baseline. This physiological buffering effect was significantly superior to that in the comfortable scenario. Subsequent interviews confirmed strong attachment and trust (e.g., "I only dared to follow its movement path"), indicating that the high-stress threat simulation successfully activated the users' attachment behavioral

system. Its immediate anxiety relief effect (reduction in STAI-S scores) outperformed simple relaxation therapies.

Generalized Anxiety Disorder (GAD) Tendency Group ($N = 3$): The interaction pattern was distinctly different. In the low-arousal comfortable environment, although the PANAS and STAI-S scales indicated a decrease in negative emotions and an increase in subjective pleasure, changes in physiological arousal were mild, lacking the dramatic physiological regulatory effects seen in the specific phobia group. Interviews revealed that these three users viewed the virtual character as an "interesting and relaxing companion" but did not develop deep emotional dependency or exclusive attachment.

This stark contrast suggests a potential "stress threshold" for establishing deep human-computer attachment relationships. In low-arousal situations lacking clear external threats, the virtual companion struggles to trigger users' deep-seated safety needs, thereby limiting its potential as a core therapeutic element. This provides a crucial empirical basis and direction for the future design of adaptive VR interventions tailored to different anxiety types (especially for GAD patients).

4 Conclusion and future works

Through empirical data, this study reveals for the first time in a VR context the critical role of environmental stress as a moderating

DEAR FRIEND SYSTEM: CONDITIONS

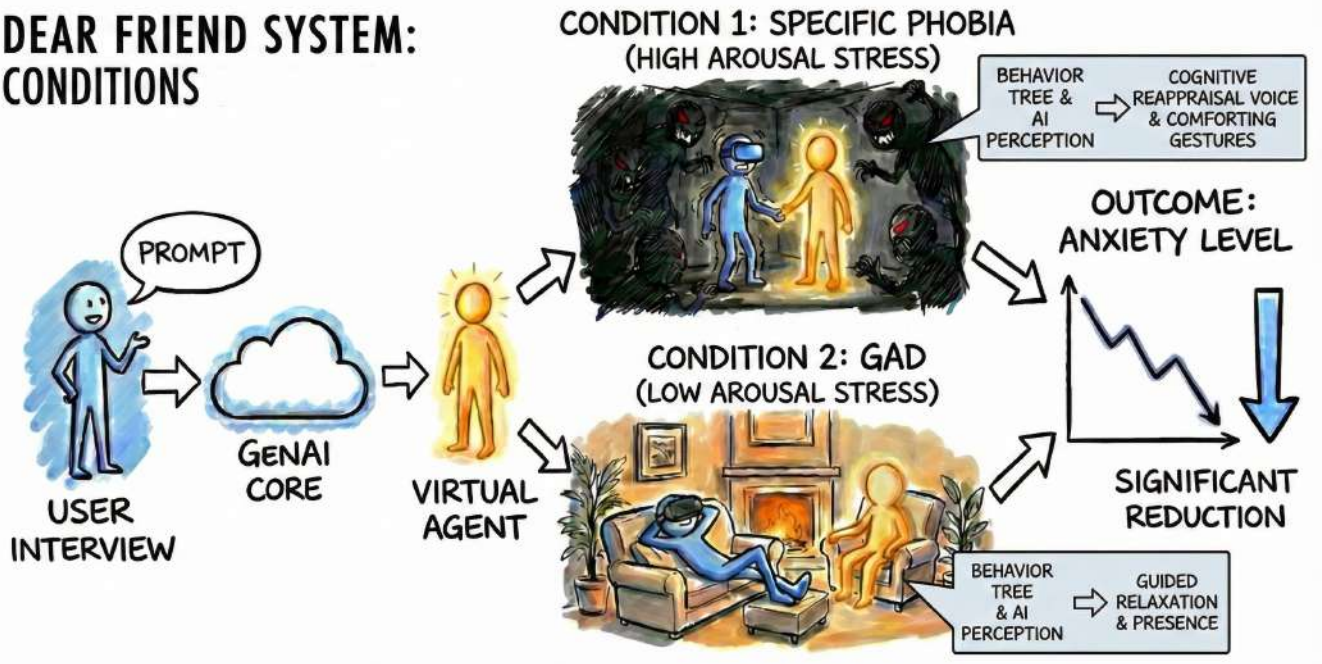


Figure 4: Dear Friend System Conditions.

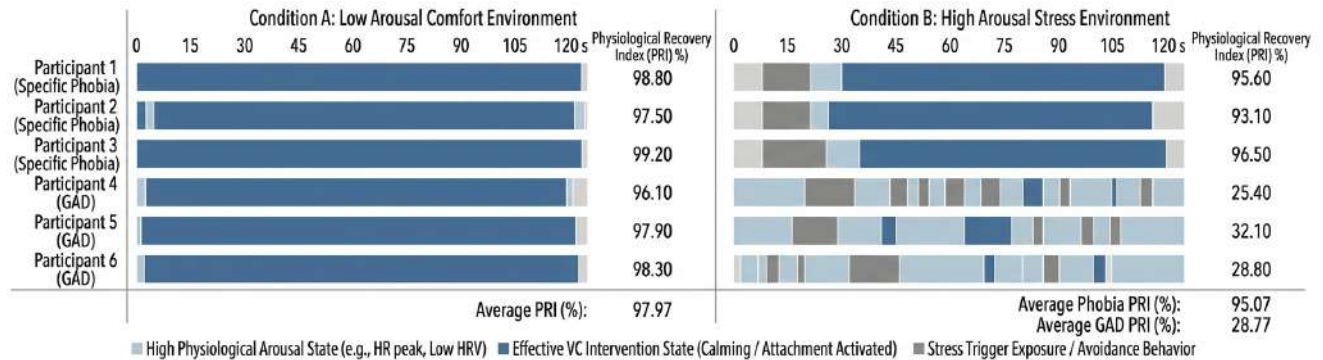


Figure 5: Comparison of User PRI Indices in Two Environments.

variable in human-machine attachment formation. Experimental results indicate that high-intensity environmental stress may be a necessary condition for activating the virtual social buffering effect: for specific phobia users, the “survival threat” simulated by the high-fidelity exposure environment forces an extreme contraction of their cognitive resources. At this time, the stable social signals provided by the virtual partner fill the huge vacuum of security, thereby catalyzing a strong emotional bond in a very short time. However, for general anxiety users, due to the lack of a specific object of fear, interactions in a comfortable environment remain more at the level of a friendship relationship and are difficult to touch upon deep emotional regulation mechanisms.

Although this pre-experiment provides valuable preliminary evidence, the limitation of the sample size ($N = 6$) requires us to

be cautious in generalizing the conclusions. Future work will be dedicated to expanding the sample size and introducing longitudinal tracking studies to explore whether this emotional attachment established quickly under high pressure has long-term therapeutic maintenance effects, and whether there are ethical risks of users developing excessive dependence on virtual images in real life. Furthermore, we plan to upgrade the system in the next stage by introducing real-time adaptive dialogue functions based on Large Language Models (LLM), enabling the virtual partner to not only provide non-verbal companionship but also conduct deep cognitive behavioral guidance based on the user’s real-time physiological feedback, thereby constructing a truly closed-loop intelligent psychological intervention system.

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